

Finite Difference Scheme for the Ornstein–Uhlenbeck process

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July 2026

Consider the following Fokker–Planck equation defined on $[x_L, x_R]$, where $x_L = -5$ and $x_R = 5$:

$$\partial_t \rho = \kappa \partial_x(x\rho) + D \partial_{xx} \rho, \quad x \in [x_L, x_R], \quad t \geq 0.$$

The probability flux is defined by

$$J(x, t) = -\kappa x \rho(x, t) - D \partial_x \rho(x, t).$$

We impose the zero-flux boundary conditions

$$J(x_L, t) = J(x_R, t) = 0, \quad t \geq 0.$$

Let $T > 0$ be the maximum time. For $M, N \in \mathbb{N}$, define

$$h = \frac{x_R - x_L}{M} = \frac{10}{M}, \quad \delta = \frac{T}{N}.$$

The spatial and temporal grid points are given by

$$x_j = x_L + jh, \quad j = 0, 1, \dots, M,$$

and

$$t^{(n)} = n\delta, \quad n = 0, 1, \dots, N.$$

We write

$$\rho_j^n := \rho(x_j, t^{(n)}).$$

At the interior grid points, the first and second spatial derivatives are approximated using central differences:

$$\partial_x(x\rho)(x_j, t^{(n)}) \approx \frac{x_{j+1}\rho_{j+1}^n - x_{j-1}\rho_{j-1}^n}{2h},$$

and

$$\partial_{xx}\rho(x_j, t^{(n)}) \approx \frac{\rho_{j+1}^n - 2\rho_j^n + \rho_{j-1}^n}{h^2}.$$

Using the forward Euler method for the time derivative, we obtain

$$\frac{\rho_j^{n+1} - \rho_j^n}{\delta} = \kappa \frac{x_{j+1}\rho_{j+1}^n - x_{j-1}\rho_{j-1}^n}{2h} + D \frac{\rho_{j+1}^n - 2\rho_j^n + \rho_{j-1}^n}{h^2}.$$

Therefore, for $j = 1, \dots, M-1$,

$$\rho_j^{n+1} = \rho_j^n + \frac{\kappa\delta}{2h} (x_{j+1}\rho_{j+1}^n - x_{j-1}\rho_{j-1}^n) + \frac{D\delta}{h^2} (\rho_{j+1}^n - 2\rho_j^n + \rho_{j-1}^n).$$

Elimination of the ghost points

To encode the boundary conditions, introduce the ghost points $x_{-1} = x_0 - h$ and $x_{M+1} = x_M + h$.

At the left boundary, the zero-flux condition gives

$$0 = -\kappa x_0 \rho_0^n - D \frac{\rho_1^n - \rho_{-1}^n}{2h}.$$

Hence,

$$D \frac{\rho_1^n - \rho_{-1}^n}{2h} = -\kappa x_0 \rho_0^n,$$

and therefore

$$\rho_{-1}^n = \rho_1^n + \frac{2\kappa h}{D} x_0 \rho_0^n.$$

Similarly, at the right boundary,

$$0 = -\kappa x_M \rho_M^n - D \frac{\rho_{M+1}^n - \rho_{M-1}^n}{2h},$$

so that

$$\rho_{M+1}^n = \rho_{M-1}^n - \frac{2\kappa h}{D} x_M \rho_M^n.$$

For notational convenience, define

$$\alpha := \frac{\kappa\delta}{2h}, \quad \beta := \frac{D\delta}{h^2}.$$

We also define the boundary coefficients

$$\begin{aligned} a_L &:= -\frac{\kappa^2\delta}{D} x_{-1} x_0, & a_R &:= -\frac{\kappa^2\delta}{D} x_{M+1} x_M, \\ b_L &:= \frac{2\kappa h}{D} x_0 - 2, & b_R &:= -\frac{2\kappa h}{D} x_M - 2. \end{aligned}$$

Let the discrete density vector be

$$\boldsymbol{\rho}^n := (\rho_0^n \quad \rho_1^n \quad \cdots \quad \rho_M^n)^\top.$$

After eliminating the ghost values, the fully discrete scheme can be written in matrix form as

$$\boldsymbol{\rho}^{n+1} = (I_{M+1} + A + B) \boldsymbol{\rho}^n,$$

where I_{M+1} is the $(M+1) \times (M+1)$ identity matrix, A represents the drift contribution, and B represents the diffusion contribution.

The drift matrix is

$$A = \begin{pmatrix} a_L & \kappa\delta & 0 & \cdots & 0 \\ -\alpha x_0 & 0 & \alpha x_2 & \ddots & \vdots \\ 0 & -\alpha x_1 & 0 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & \alpha x_M \\ 0 & \cdots & 0 & \kappa\delta & a_R \end{pmatrix}.$$

The diffusion matrix is

$$B = \beta \begin{pmatrix} b_L & 2 & 0 & \cdots & 0 \\ 1 & -2 & 1 & \ddots & \vdots \\ 0 & 1 & -2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & 1 \\ 0 & \cdots & 0 & 2 & b_R \end{pmatrix}.$$

Thus, the numerical solution is updated at each time step according to

$$\boxed{\boldsymbol{\rho}^{n+1} = (I_{M+1} + A + B) \boldsymbol{\rho}^n}.$$